ASSIGNMENT-2 : CONDITIONAL STATEMENTS

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GROUP-E1, ROLL NO:7

1. Find the greatest among three numbers.

In [3]:
```
a = int(input('Enter a: '))
b = int(input('Enter b: '))
c= int(input('Enter c: '))
print('Greatest among these is: ', end='')
if (a>b):
    if(a>c):
        print(a)
    else:
        print(c)
else:
    if(b>c):
        print(b)
    else:
        print(c)
```

```
Enter a: 4
Enter b: 2
Enter c: 8
Greatest among these is: 8
```

2. Asks the user to enter a length in centimetres. If the user enters a negative length, the program should tell the user that the entry is invalid. Otherwise, the program should convert the length to inches and print out the result. (1 inch = 2.54 cm)

In [6]:
```
l_cm = int(input('Enter length in centimeters: '))
if l_cm<0 :
    print('Invalid length.')
else:
    l_inch = l_cm/2.54
    print(f'Length in inches: {l_inch:.2f} inch')
```

```
Enter length in centimeters: 150
Length in inches: 59.06 inch
```

3. Ask the user for a temperature. Then ask them what units, Celsius or Fahrenheit, the temperature is in. Your program should convert the temperature to the other unit. The conversions are F = 9/5C +32 and C = 5/9(F-32).

In [23]:
```
t = float(input('Enter temperature: '))
d = int(input('If temperature is in degree Celcius, enter 1 \nIf temperature is in degree Fa
if d==1 :
    f = 9/5*t + 32
    print('Temperature in Fahrenheit scale: ', f'{f:.2f}')
elif d==2 :
    c = 5/9 * (t-32)
    print('Temperature in Celcius scale: ', f'{c:.2f}')
else :
    print('Invalid input!!!')
```

```
Enter temperature: 104
If temperature is in degree Celcius, enter 1
If temperature is in degree Fahrenheit, Enter 2 :
2
Temperature in Celcius scale:  40.00
```

4. Calculate electric bill as follows: for first 100 units Rs 1.50 per unit, for next 100 units Rs 3.75 per unit, for next 100 units Rs 4.25, for rest units Rs 5.65 per unit.

In [29]:
```
u = int(input('Enter number of units : '))
if u<0:
    print('Invalid input!!!')
else:
    if u<=100:
        e = u*1.5
    elif u<=200:
        e = (u-100)*3.75 + 100*1.5
    elif u<=300:
        e = (u-200)*4.25 + 100*3.75 + 100*1.5
    else:
        e = (u-300)*5.65 + 100*4.25 + 100*3.75 + 100*1.5
    print('Electric bill : ', e)
```

```
Enter number of units : 125
Electric bill :  243.75
```

5. Ask the user to enter a temperature in Celsius. The program should print a message based on the temperature:

If the temperature is less than -273.15, print that the temperature is invalid because it is below absolute zero.

If it is exactly -273.15, print that the temperature is absolute 0.

If temperature is between -273.15 and 0, print temperature is below freezing.

If it is 0, print that the temperature is at the freezing point.

If it is between 0 and 100, print the temperature is in the normal range.

If it is 100, print that the temperature is at the boiling point.

If it is above 100, print that the temperature is above the boiling point

In [36]:
```
c = float(input('Enter temperature in celcius: '))
if c<-273.15 :
    print('Invalid temperature - below absolute 0.')
elif c == -273.15 :
    print('Absolute 0 temperature.')
elif c>-273.15 and c<0:
    print('Below freezing temperature.')
elif c==0:
    print('Freezing point temperature.')
elif c>0 and c<100:
    print('Normal range temperature.')
else :
    print('Above boiling point temperature.')
```

```
Enter temperature in celcius: 24
Normal range temperature.
```

6. A store charges Rs. 12 per item if you buy less than 10 items. If you buy between 10 and 99 items, the cost is Rs. 10 per item. If you buy 100 or more items, the cost is

Rs. 7 per item. Write a program that asks the user how many items they are buying and prints the total cost.

In [38]:
```
i = int(input('Enter number of items: '))
if i<0:
    print('Invalid input!!!')
else:
    if i<10:
        c = i*12
    elif i<=99:
        c = i*10
    else:
        c=i*7
    print(f'Total cost : Rs. {c}')
```

```
Enter number of items: 2
Total cost : Rs. 24
```

7. A year is a leap year if it is divisible by 4, except that years divisible by 100 are not leap years unless they are also divisible by 400. Write a program that asks the user for a year and prints out whether it is a leap year or not.

In [43]:
```
y = int(input('Enter a year: '))
if y%4==0 and y%100!=0 or y%400==0:
    print('Leap year.')
else:
    print('Not a Leap Year.')
```

```
Enter a year: 1900
Not a Leap Year.
```